Aishvarya Salvi

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Summary

AI/ML Engineer building production-grade LLM applications, multi-agent systems, and forecasting pipelines using Anthropic Claude, OpenAI, LangChain/LangGraph, and Python. Delivers end-to-end AI solutions — from data pipelines to FastAPI services to stakeholder-facing dashboards with measurable impact

Education

University at Buffalo, Master of Science in Data Science 2025
Pune University, Bachelor's in Computer Science 2021

Professional Experience

AI Engineer, Mediaocean 02/26 – Present

- Owned end-to-end design and deployment of two AI-powered systems — a regression test-selection optimizer and a RAG-based failure classifier — replacing manual full-suite QA across 100+ repos and 15k+ tests.
- Evaluated Oracle AI Vector Search vs. pgvector for production embeddings and drove adoption of pgvector, then validated the ranking approach against live Jira data — uncovering a component-mapping flaw affecting 20k+ tests.
- Engineered a five-stage multi-agent RAG classifier (pgvector, Ollama) with an LLM-free fast-path and deterministic decision gate, cutting classification latency 90x.
- Containerized both systems (Docker, docker-compose, nginx) with CI/CD automation and an async, rate-limit-aware architecture keeping pipeline runtime under 60 seconds.
- Collaborated with Automation, Engineering, and Data Science leadership in Agile/Scrum, shipped LLM based stakeholder tools, driving a 60% reduction in manual QA effort.

AI/ML Engineer, New Era Cap 01/25 – 01/26

- Built and deployed an end-to-end demand forecasting pipeline on Azure/Snowflake, replacing manual Excel-based purchase and vendor planning to reduce overstocking and stockouts.
- Migrated and validated hundreds of Excel sources into structured Snowflake tables, then performed EDA, cleaning, and time-series feature engineering to inform model design.
- Benchmarked multiple forecasting models (XGBoost, LightGBM, ARIMA, SARIMA, CatBoost), selecting CatBoost for 92% accuracy, and deployed it to production on Azure.
- Built a Power BI dashboard and a GenAI-powered Streamlit app (FastAPI + OpenAI API) translating SHAP feature importance into natural-language explanations for non-technical stakeholders.

Senior Software Engineer, Mediaocean 08/20 – 07/24

- Built API and UI test automation frameworks (Karate, Selenium/POM) integrated into Dockerized Jenkins CI/CD pipelines, validating campaign workflows for Mediaocean's Prisma platform.
- Engineered SQL-based validation and Power BI reporting, improving test coverage from 60% to 95% and cutting escaped defects by 40% and resolution time by 30%.

Technical Skills

- **Programming & Frameworks:** Python, SQL, JavaScript/TypeScript, FastAPI, REST APIs, React
- **AI & LLM Engineering:** LLM integration (Anthropic Claude, OpenAI, Ollama), prompt engineering, agentic workflows, LangChain, LangGraph, multi-agent orchestration, RAG, vector databases (Pinecone, pgvector), semantic search
- **ML & Data Science:** Forecasting & regression models (XGBoost, LightGBM, CatBoost, ARIMA, SARIMA), classification, model evaluation, feature engineering, explainable AI (SHAP), Apache Airflow, Jenkins
- **Databases & Data Warehousing:** PostgreSQL, Snowflake, S3
- **Cloud & DevOps:** AWS (EC2, S3), Azure (Data Factory, Blob Storage), Docker/docker-compose, nginx, CI/CD
- **Data Engineering & Visualization:** ETL/ELT, SQL-based validation, Power BI, Streamlit, Matplotlib/Seaborn

Projects

Multi-Agent RAG Pipeline for Automated Regression Failure Triage

- Built a multi-agent RAG pipeline that auto-classifies scipt failures using an LLM based fast-path to cut classification latency 90% on high-confidence cases.

AI-Powered Multi-Pipeline Regression Test Recommendation Platform

- Shipped a dual-pipeline platform (FastAPI/React) that auto-recommends regression tests from Bitbucket commits using AI, replacing manual QA triage upto 60%.

Multi-Agent Supply Chain Disruption Detection & Risk Monitoring Platform

- Built a LangGraph multi-agent system (Pinecone, Airflow, LangChain) that scores supplier risk and recommends validated alternative suppliers in real time.